DESIGN NOTES — KEEP KNOXVILLE BEAUTIFUL

System scope

This logical design supports volunteer event coordination and inventory management. It includes volunteers, events, recurring locations, event participation, supplies, supply requests, and inventory transactions.

Relationships

- LOCATION to EVENT is one-to-many: each event uses one location, while a location may host zero or many events.
- VOLUNTEER to EVENT is many-to-many. EVENT_PARTICIPATION resolves it with the composite primary key (event_id, volunteer_id). The junction stores role, hours_worked, and registered_at because these facts belong to a volunteer's participation in a particular event.
- EVENT to SUPPLY is many-to-many through SUPPLY_REQUEST. A request stores quantity_requested, request_status, and requested_at, which are specific to a supply request for an event.
- SUPPLY to INVENTORY_TRANSACTION is one-to-many: each transaction records a change for one supply, while a supply can have many restock, usage, or adjustment records. An inventory transaction may optionally reference an event when the change is event-related.

Normalization

Volunteer contact information is stored once in VOLUNTEER rather than repeated on every participation row. Event participation stores only the relationship-specific facts. Similarly, supply descriptions and reorder levels are stored once in SUPPLY instead of being copied into every request or transaction. This reduces repeated data and prevents update and inconsistency problems.

Important design decisions

EVENT_PARTICIPATION uses a composite primary key so the same volunteer cannot be registered twice for the same event. Surrogate integer keys are used for the main entities and transaction/request records because they provide stable, simple references in SQLite. Status and transaction_type are stored as attributes for the initial logical design; the implementation can later enforce allowed values with application validation or CHECK constraints.

The event-to-inventory-transaction relationship is optional because restocking and general adjustments may occur without being tied to a specific event. The location entity is separate from EVENT so recurring service locations are not duplicated across events.

Changes from the proposal

The design refines the proposed concepts without changing the approved scope. LOCATION was included as a meaningful supporting entity, and the proposal's supply requests and inventory activity were made explicit through SUPPLY_REQUEST and INVENTORY_TRANSACTION. The volunteer/event many-to-many relationship was implemented through EVENT_PARTICIPATION as planned.